

Phase I/II Operating Characteristics

TITE cycle 2 and cycle 3 versus multicycle additive

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%) - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	9.3	5.7	3.4	1.2	0.4	80
C2 multicycle one-stage	8.5	5.7	2.9	1.7	0.2	81
C2 TITE two-stage	9.4	6.6	3.2	1.5	0	79.3
C2 multicycle two-stage	8.8	6	2.8	1.5	0.2	80.7
C3 TITE one-stage	8.8	6.8	4.1	1	0.3	79
C3 multicycle one-stage	9.2	7.5	3.3	1.4	0.1	78.5
C3 TITE two-stage	9.4	6.7	4	1	0.1	78.8
C3 multicycle two-stage	10.5	6.6	3.9	1.2	0.2	77.6
Mean administrations by dose - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	11.1	4.7	2.5	1	0.3	
C2 multicycle one-stage	10.6	4.8	2.8	1.2	0.4	
C2 TITE two-stage	11	4.8	2.6	1	0.4	
C2 multicycle two-stage	10.6	4.8	2.8	1.2	0.4	
C3 TITE one-stage	11.4	4.7	2.7	1	0.3	
C3 multicycle one-stage	10.8	4.7	2.6	1.1	0.4	
C3 TITE two-stage	11.2	4.8	2.8	1.1	0.3	
C3 multicycle two-stage	10.8	4.7	2.7	1.2	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%) - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	24.3	21.9	21.4	16.4	15.3	0.7
C2 multicycle one-stage	23.5	22.4	19.8	18	14.8	1.5
C2 TITE two-stage	23.8	21.9	20.7	18.1	14.8	0.7
C2 multicycle two-stage	22.3	22.5	18.6	19.6	15.6	1.4
C3 TITE one-stage	23	24	20.3	18.3	13	1.4
C3 multicycle one-stage	24.3	21.6	18.8	19.5	14.7	1.1
C3 TITE two-stage	22.9	24	20.6	18	13.2	1.3
C3 multicycle two-stage	23.7	22	17.6	19.8	15.8	1.1
Mean administrations by dose - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	7.1	6.5	5.9	5.5	4.9	
C2 multicycle one-stage	7.1	6.6	5.9	5.5	4.9	
C2 TITE two-stage	6.9	6.5	6	5.6	5.1	
C2 multicycle two-stage	6.9	6.4	6	5.7	5	
C3 TITE one-stage	6.9	6.7	6	5.6	4.8	
C3 multicycle one-stage	7	6.4	5.7	5.9	4.9	
C3 TITE two-stage	6.8	6.7	6	5.4	5	
C3 multicycle two-stage	6.9	6.4	5.7	6	5.1	

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Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%) - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	4	45.8	28.7	12.9	3.1	5.5
C2 multicycle one-stage	4.6	44.1	29.6	13.4	3	5.3
C2 TITE two-stage	5.5	45.4	29.4	11.6	3.1	5
C2 multicycle two-stage	5.1	45	30.1	11.5	2.9	5.4
C3 TITE one-stage	4	45	30.5	12.6	2.2	5.7
C3 multicycle one-stage	3.6	46.5	28.8	13.1	3.2	4.8
C3 TITE two-stage	4.2	44.4	30.1	11.5	2.9	6.9
C3 multicycle two-stage	4.7	47.4	27.9	12.6	1.8	5.6
Mean administrations by dose - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	5.2	10.9	7.6	4.4	1.5	
C2 multicycle one-stage	5.2	10.7	7.8	4.3	1.7	
C2 TITE two-stage	5.1	10.2	8	4.7	1.7	
C2 multicycle two-stage	5	10.2	8	4.7	1.9	
C3 TITE one-stage	5.2	10.9	7.7	4.3	1.5	
C3 multicycle one-stage	5.2	10.6	7.8	4.3	1.7	
C3 TITE two-stage	5.1	10.2	8	4.6	1.8	
C3 multicycle two-stage	5.1	10.2	8	4.5	1.9	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%) - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	0.9	0.7	5	59.4	33.9	0.1
C2 multicycle one-stage	0.9	1	5.1	60.2	32.7	0.1
C2 TITE two-stage	1.2	0.8	5.8	53.8	38.3	0.1
C2 multicycle two-stage	1	0.9	5.7	55.4	36.9	0.1
C3 TITE one-stage	0.5	1.8	5.4	56.7	35.5	0.1
C3 multicycle one-stage	0.6	1.7	6.3	58.1	33.2	0.1
C3 TITE two-stage	0.8	1.7	7	52	38.4	0.1
C3 multicycle two-stage	0.2	1.9	5.2	59.6	33.1	0
Mean administrations by dose - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	3.4	3.5	4.5	12.2	6.5	
C2 multicycle one-stage	3.5	3.4	4.5	12.2	6.4	
C2 TITE two-stage	3.4	3.5	4.3	10.6	8.3	
C2 multicycle two-stage	3.4	3.4	4.5	10.6	8	
C3 TITE one-stage	3.4	3.5	4.6	11.7	6.8	
C3 multicycle one-stage	3.4	3.5	4.5	12	6.5	
C3 TITE two-stage	3.4	3.4	4.5	10.2	8.4	
C3 multicycle two-stage	3.4	3.4	4.3	11.2	7.7	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%) - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	9	25.3	46.7	11.1	1	6.9
C2 multicycle one-stage	11.3	21.9	48.3	11.7	0.5	6.3
C2 TITE two-stage	10.7	24.1	46.7	9.9	1.2	7.4
C2 multicycle two-stage	10.7	23.6	49.3	9.3	1	6.1
C3 TITE one-stage	9.4	22.7	49	11.2	1.3	6.4
C3 multicycle one-stage	10.1	22.8	48.1	11.4	1.2	6.4
C3 TITE two-stage	10.2	25.1	46.5	10.6	1	6.6
C3 multicycle two-stage	9.7	21.9	50.8	9.9	1.1	6.6
Mean administrations by dose - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	7.1	8.1	9.8	3.6	0.7	
C2 multicycle one-stage	7	7.8	10.3	3.5	0.6	
C2 TITE two-stage	6.9	7.8	9.5	4	1	
C2 multicycle two-stage	6.8	7.7	9.9	3.9	1	
C3 TITE one-stage	6.9	7.8	10	3.9	0.8	
C3 multicycle one-stage	7	7.8	10.1	3.8	0.6	
C3 TITE two-stage	6.7	7.7	9.4	4.4	1.2	
C3 multicycle two-stage	6.7	7.6	9.7	4.2	1	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%) - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	4	7.3	8.6	12.4	63.8	3.9
C2 multicycle one-stage	4.8	5.6	9.5	11.9	64.2	4
C2 TITE two-stage	3.9	6.8	8.1	12.9	64.4	3.9
C2 multicycle two-stage	5.2	5.3	10.1	11	64.1	4.3
C3 TITE one-stage	4.9	7	9.3	9.2	65.8	3.8
C3 multicycle one-stage	4.2	5.5	8.2	12.3	66.2	3.6
C3 TITE two-stage	4.6	6.9	9	10.2	65.3	4
C3 multicycle two-stage	4.9	5.5	8.5	12.5	65	3.6
Mean administrations by dose - cycle 2 rows, then cycle 3 rows						
C2 TITE one-stage	4.6	4.6	5.1	5.3	10.3	
C2 multicycle one-stage	4.7	4.3	5.2	5.4	10.2	
C2 TITE two-stage	4.6	4.6	5	5.3	10.3	
C2 multicycle two-stage	4.7	4.3	5.2	5.4	10.3	
C3 TITE one-stage	4.6	4.6	5	5.1	10.6	
C3 multicycle one-stage	4.7	4.4	4.9	5.3	10.6	
C3 TITE two-stage	4.5	4.6	4.9	5.2	10.7	
C3 multicycle two-stage	4.6	4.3	4.9	5.4	10.7	

Mean Unique Patients by Method

Mean number of unique patients per simulated trial

Scenario	1	16	20	24	27	38	Mean
C2 TITE one-stage	12.5	17.9	18.8	19.4	19.2	18.7	17.8
C2 multicycle one-stage	12.7	18.2	18.9	19.5	19.5	18.9	18
C2 TITE two-stage	12.6	17.9	18.9	19.6	19.4	18.7	17.8
C2 multicycle two-stage	12.7	18.2	19.1	19.6	19.6	18.9	18
C3 TITE one-stage	10.7	14.2	15.6	16.3	16.6	15.3	14.8
C3 multicycle one-stage	10.6	14.4	15.7	16.5	16.5	15.5	14.9
C3 TITE two-stage	10.7	14.2	15.7	16.4	16.8	15.4	14.8
C3 multicycle two-stage	10.7	14.5	15.9	16.7	16.7	15.6	15

Values are reported from Total_Unique_Patients and shown to one decimal. The Mean column is the unweighted mean across the six scenarios.

Rows are grouped by cycle (C2, then C3); within each cycle, one-stage rows precede two-stage rows.